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Indoor Air Quality Management Using Hydroponic Systems: A Survey Paper

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Abstract: Indoor air quality (IAQ) significantly impacts human health, with pollutants such as carbon dioxide (CO₂) and volatile organic compounds (VOCs) contributing to respiratory and other diseases. Traditional methods of improving IAQ, such as air purifiers, can be costly and energy intensive. This paper explores an innovative solution combining hydroponics and Internet of Things (IoT) technologies to improve IAQ. Using an ESP32 S3 microcontroller, the system integrates sensors to monitor CO₂ levels, temperature, humidity, and other environmental factors, while controlling actuators like LED grow lights and nutrient pumps for hydroponic plants known for their air-purifying properties. Real-time data from these sensors allows for adaptive control, ensuring optimal plant growth and pollutant reduction. Initial results show significant reductions in CO₂ levels, suggesting that this IoT-integrated hydroponic system is a cost-effective and sustainable method to enhance IAQ. The approach has potential for smart home integration and broader applications in urban air quality management.

Keywords: Indoor air quality, hydroponics, IoT, ESP32 S3, CO₂ reduction, air purification, smart home, environmental sensors, pollutant monitoring, sustainable solutions.

I. INTRODUCTION

Indoor air quality (IAQ) has emerged as a critical factor affecting human health, particularly in urban settings where air pollution is a significant concern. Poor IAQ can lead to a variety of health issues, including respiratory diseases, cardiovascular conditions, and general discomfort. The World Health Organization (WHO) has highlighted the growing importance of managing IAQ, as exposure to pollutants such as particulate matter, volatile organic compounds (VOCs), and carbon dioxide (CO₂) can have detrimental effects on both physical and mental well-being [1]. Among these, CO₂, often a product of human activity and confined spaces, poses a serious challenge. Over time, its accumulation in indoor spaces leads to discomfort, fatigue, and in some cases, more severe health consequences [2]. As a solution, various technologies have been proposed for mitigating indoor pollution. The integration of the Internet of Things (IoT) has offered innovative ways to monitor and manage IAQ more efficiently. IoT-based systems, which connect sensors, actuators, and microcontrollers, allow for real-time monitoring of indoor conditions, such as temperature, humidity, CO₂ levels, and air purity. These systems are capable of making adaptive adjustments based on the collected data, which can lead to more effective pollutant removal and improved overall IAQ [3].

One promising approach for enhancing IAQ is combining hydroponics with IoT. Hydroponics, the soil-less cultivation of plants, has been shown to be an effective method for purifying air indoors. Certain plant species, such as Peace Lily and Areca Palm, are known to absorb toxins, including CO₂, effectively from the surrounding environment [5][6]. By integrating hydroponics with IoT technology, it becomes possible to optimize plant growth, improve air purification efficiency, and maintain optimal conditions for both plant health and air quality. A microcontroller like the ESP32 S3 can serve as the central hub for managing sensor data and controlling actuators, ensuring that the system adapts in real-time to changes in environmental conditions [7].

This paper explores the potential of integrating IoT technology with hydroponic systems to address IAQ challenges. By focusing on the use of an ESP32 S3 microcontroller and associated sensors, the research aims to develop a cost-effective solution that can efficiently monitor and regulate IAQ, particularly in spaces with high CO₂ concentrations and other common indoor pollutants [8]. The study will investigate how this system can be used for improving IAQ in indoor environments, with the hope of providing an efficient, scalable solution that can be integrated into homes, offices, and other indoor spaces.

II. BACKGROUND AND RELATED WORK

The relationship between plants and air quality has been well-documented in scientific research, particularly by NASA's Clean Air Study, which identified several plant species capable of removing harmful pollutants from indoor environments. Peace Lily (*Spathiphyllum*) and Areca Palm (*Dyopsis lutescens*) are two such plants known for their ability to absorb CO₂ and VOCs, including benzene, formaldehyde, and trichloroethylene, which are commonly found in indoor spaces [5][6].

This discovery has led to increased interest in using plants as natural air purifiers in homes, offices, and other indoor environments. Hydroponics, a method of growing plants without soil by providing them with a nutrient-rich water solution, further enhances this process. Hydroponic systems are advantageous because they allow for more controlled and efficient plant growth, which can lead to faster pollutant absorption rates. Traditional soil-based growing methods often suffer from inconsistencies in nutrient supply and water retention, but hydroponics eliminates these limitations, offering a more sustainable and efficient solution for both plant cultivation and air purification [9].

While the use of plants for improving air quality has been widely researched, the integration of hydroponics with IoT is a relatively new area of study. The application of IoT technologies to agriculture has been widely explored, particularly in the context of precision farming. By integrating sensors, actuators, and microcontrollers, IoT systems enable real-time monitoring and automation of various environmental factors, including temperature, humidity, light intensity, and nutrient levels. These systems can optimize conditions for plant growth, resulting in healthier plants that can absorb pollutants more effectively [4][6]. However, the application of IoT in indoor air quality management, specifically in hydroponic systems, is still in its early stages, with only a handful of studies exploring the potential for this combination.

Several studies have investigated the use of IoT systems to monitor and improve IAQ. For example, Liu et al. (2019) demonstrated how IoT-enabled devices can continuously track indoor air pollutants such as CO₂, particulate matter, and VOCs. These systems can trigger automatic actions, such as activating ventilation systems or air purifiers, when pollutant levels exceed certain thresholds [7]. Similarly, Singh et al. (2022) explored the use of IoT systems in air quality management, focusing on real-time data collection and adaptive control strategies to mitigate indoor air pollution [8]. However, while these studies highlight the potential of IoT for IAQ management, they do not specifically explore the integration of hydroponics with IoT, which could enhance both air purification and plant health.

The integration of hydroponics with IoT offers several advantages over traditional methods. First, it enables real-time monitoring of environmental parameters, such as CO₂ levels, temperature, humidity, and light intensity, which can directly impact plant growth and pollutant absorption. Second, IoT systems can automate actions such as adjusting nutrient delivery, activating LED grow lights, or regulating water flow, ensuring optimal conditions for both plants and IAQ. This level of automation and control is difficult to achieve with traditional methods of plant cultivation, which rely on manual intervention and often lack the precision necessary for consistent results.

Despite the promising potential of IoT-integrated hydroponic systems for IAQ management, there are several challenges that need to be addressed. These include the calibration of sensors for accurate pollutant measurement, the maintenance of system components, and the high initial cost of setting up such systems. Nonetheless, as IoT technology continues to evolve and become more affordable, the integration of hydroponics with IoT holds significant promise for improving indoor air quality in both residential and commercial settings [7][8].

III. RESEARCH GAP ANALYSIS AND OBJECTIVES

While hydroponics has been extensively utilized in agriculture for food production, its application in managing indoor air quality (IAQ) has not been fully explored. The existing body of research primarily focuses on hydroponic systems for enhancing plant growth, nutrient delivery, and food security, with limited emphasis on their potential role in mitigating indoor pollutants such as carbon dioxide (CO₂) and volatile organic compounds (VOCs). Although plants have been recognized for their air-purifying capabilities, and hydroponic systems have demonstrated efficient plant growth, a gap exists in understanding how these systems can be optimized for IAQ management.

The integration of IoT in hydroponics offers a promising solution to address this gap. Real-time monitoring and automation can enhance the efficiency of hydroponic systems by continuously measuring and adjusting environmental parameters. However, despite the advancement of IoT applications in agriculture and environmental monitoring, few studies have focused on combining hydroponics with IoT for IAQ management. Most research on IAQ has concentrated on traditional air purification systems such as air purifiers and ventilation systems, which do not provide the added benefits of plant-based purification [10][11]. Thus, there is an opportunity to bridge this gap by investigating the effectiveness of IoT-integrated hydroponic systems specifically for IAQ improvement.

One of the key challenges in existing systems is the lack of real-time data and adaptive control mechanisms. Traditional IAQ management systems are typically passive and require manual intervention when pollutant levels exceed acceptable limits. In contrast, IoT-integrated systems can actively monitor pollutants and adjust environmental factors such as light, humidity, and nutrient delivery to optimize plant health and air purification efficiency.

Additionally, these systems can provide valuable insights into the long-term effectiveness of plant-based purification, enabling the identification of specific plants or hydroponic setups that perform best in reducing IAQ pollutants [8][9].

Another significant gap is the cost and accessibility of such systems. Many of the existing IoT-based IAQ solutions are expensive and may not be easily accessible to a wide range of users, especially in residential settings. This paper aims to address this by exploring cost-effective solutions that leverage affordable components such as the ESP32 S3 microcontroller and low-cost sensors. These technologies, when integrated into a hydroponic system, could provide an affordable and scalable solution for IAQ management in homes, offices, and other indoor spaces [12].

The primary objective of this research is to investigate the feasibility and effectiveness of a cost-effective, IoT-integrated hydroponic system for improving IAQ. By incorporating real-time monitoring and adaptive control, the system will aim to reduce indoor pollutants like CO₂ and VOCs, leveraging the natural air-purifying capabilities of plants. The objectives include:

- Developing a system that integrates IoT-based sensors and hydroponics to optimize IAQ.
- Testing the effectiveness of different plant species in reducing specific pollutants (e.g., CO₂, VOCs).
- Providing a cost-effective solution that can be widely implemented in indoor environments.

Ultimately, this paper seeks to fill the research gap in IAQ management by developing a practical and scalable solution that combines the benefits of hydroponics and IoT, offering new possibilities for sustainable and efficient air quality management in indoor environments.

IV. HYDROPONICS FOR INDOOR AIR QUALITY MANAGEMENT

Hydroponics is a method of growing plants in a water-based, nutrient-rich solution, rather than in soil. This system offers several advantages over traditional soil-based farming, such as faster plant growth, more efficient use of water and nutrients, and the ability to grow in various environments. These characteristics make hydroponics an appealing solution for improving indoor air quality (IAQ). Research has consistently shown that certain plants can absorb indoor air pollutants such as carbon dioxide (CO₂) and volatile organic compounds (VOCs), which are harmful to human health [5][6]. When combined with the benefits of hydroponics, these plants can provide an effective and sustainable means of purifying indoor air.

Plants, particularly those identified in studies such as the NASA Clean Air Study, have been shown to absorb not only CO₂ but also various toxic substances like formaldehyde, benzene, and trichloroethylene. These substances are commonly found in indoor environments and are associated with health issues such as respiratory problems, headaches, and even long-term chronic diseases [5]. Among the plants noted for their air-purifying properties are the Peace Lily (*Spathiphyllum*) and the Areca Palm (*Dypsis lutescens*), which are particularly effective in absorbing toxins and purifying indoor air [13][14]. When grown in a hydroponic setup, these plants can thrive without the limitations of soil, allowing them to better absorb pollutants from the surrounding environment.

The integration of hydroponic systems with real-time monitoring tools enhances the potential of these plants to effectively reduce indoor air pollutants. By incorporating sensors into the hydroponic setup, such as CO₂ sensors, temperature/humidity sensors, and TDS (Total Dissolved Solids) sensors, it becomes possible to continuously monitor and adjust the environmental conditions in the system to optimize plant growth and air purification. The ability to measure CO₂ levels and other critical factors in real-time ensures that both plant health and pollutant reduction are maximized [16]. These sensors allow for precise control of the system, adjusting light, humidity, and nutrient levels to support the optimal growth of air-purifying plants, which in turn enhances their ability to remove pollutants from the air.

Moreover, hydroponics offers a more controlled environment for plants, as the growth medium (nutrient-enriched water) provides consistent and regulated access to nutrients, ensuring healthier and faster plant growth compared to traditional soil-based agriculture. In indoor environments where air quality can fluctuate due to various factors, hydroponics allows for constant optimization of conditions such as light, temperature, and moisture, all of which play a role in the efficiency of pollutant absorption [15][16]. Additionally, because hydroponic systems do not require soil, they reduce the risk of allergens and dust, further contributing to improved air quality in indoor spaces.

While traditional air purification methods, such as air purifiers, can remove particulate matter and some pollutants from the air, they do not offer the added benefits of environmental enhancement provided by plants. Hydroponic systems not only purify the air but also contribute to aesthetic value, creating a more pleasant and natural indoor environment. In addition, the integration of IoT with hydroponics allows for greater precision and adaptability in maintaining optimal conditions for both plant growth and pollutant removal.

The combination of hydroponics and IoT opens the door to scalable and sustainable air quality solutions. For instance, in urban areas where outdoor air quality is often compromised, hydroponic systems can be used in homes, offices, or even commercial spaces to reduce the concentration of indoor pollutants. This approach provides a more sustainable alternative to conventional air purifiers, which rely on filters that need to be regularly replaced, contributing to waste and additional costs. The IoT-enabled hydroponic systems can continuously adjust to changes in pollutant levels, ensuring that the air remains clean and healthy for occupants without the need for constant manual intervention.

In conclusion, hydroponics offers a promising and sustainable solution for managing indoor air quality. By integrating IoT-based sensors and optimizing the environmental conditions for plant growth, hydroponic systems can effectively enhance air purification and contribute to healthier indoor environments. As research into this field progresses, we may see more widespread adoption of these systems as a cost-effective and eco-friendly alternative to traditional air purification methods.

V. METHODOLOGY

The methodology for integrating hydroponics with IoT for indoor air quality (IAQ) management revolves around the development and implementation of a smart hydroponic system capable of real-time monitoring and control. This approach combines plant-based air purification with advanced sensors and actuators, using an ESP32 S3 microcontroller as the central control unit. The system is designed to optimize plant growth while maintaining a healthy indoor air environment by addressing pollutants such as CO₂ and VOCs.

A. System Design and Components

The proposed system is built around several key components that work in unison to monitor and regulate both the environmental factors affecting plant growth and the air quality parameters within the indoor space. These components include:

1) Sensors:

- o CO₂ Sensor: This sensor continuously monitors the concentration of carbon dioxide in the room. Elevated CO₂ levels are a major indicator of poor indoor air quality, and by tracking them, the system can determine when air purification is necessary. The CO₂ sensor will relay data to the ESP32 S3, which can then trigger actions such as activating ventilation or adjusting the environmental settings to improve IAQ.
- o TDS Sensor: The Total Dissolved Solids (TDS) sensor monitors the concentration of nutrients in the water solution used by the hydroponic system. This sensor ensures that plants receive the correct nutrient mix to promote optimal growth, which is crucial for enhancing their pollutant absorption capacity [18]. The TDS readings are sent to the ESP32 S3 for analysis, helping the system maintain the appropriate nutrient balance.
- o Temperature and Humidity Sensors: These sensors track the indoor temperature and humidity, which are critical for maintaining plant health. Temperature and humidity levels directly impact the rate of photosynthesis and transpiration in plants, which in turn affects their ability to absorb CO₂ and other toxins. By monitoring these factors in real-time, the system can ensure that the environment is conducive to both plant growth and IAQ management [17][19].

2) Actuators:

- o LED Grow Lights: Plants in a hydroponic system require appropriate lighting for photosynthesis. LED grow lights are used in the proposed system to simulate optimal natural light conditions, ensuring that the plants grow healthily and efficiently. The light intensity and duration are controlled based on the real-time feedback from the sensors, ensuring that the plants receive the right amount of light to thrive and purify the air [19].
- o Peristaltic Pumps: These pumps are used to deliver nutrient solutions to the hydroponic system, ensuring that the plants receive a consistent supply of nutrients. The pumps are controlled based on the TDS sensor readings, adjusting the flow of nutrients to maintain the ideal concentration levels for plant growth [20].

B. Control Hub: ESP32 S3 Microcontroller

The ESP32 S3 microcontroller serves as the heart of the system, processing data from all sensors and actuators to maintain optimal plant growth conditions and IAQ. The ESP32 S3, with its dual-core processor, is well-suited for real-time applications due to its high processing power and wireless connectivity features. It facilitates the communication between the sensors, actuators, and the user interface, allowing for remote monitoring and control via a smartphone or computer. This enables users to adjust the system's settings and track air quality metrics without needing to be physically present at the installation site.

The microcontroller processes sensor data in real-time to make decisions regarding the activation of actuators. For example, if the CO₂ sensor detects a spike in CO₂ levels, the ESP32 S3 will trigger the necessary actions, such as increasing the water flow to the plants, adjusting the light intensity, or activating additional air purifiers or ventilation systems, if included in the setup. This real-time control is essential for maintaining an environment conducive to both healthy plant growth and clean indoor air.

C. Real-Time Data Analysis and Adaptive Control

One of the key aspects of this methodology is the real-time analysis of data from the sensors. The system continuously evaluates the data from the CO₂, TDS, and temperature/humidity sensors and adjusts the environmental factors as necessary. For example, if CO₂ levels rise above a set threshold, the system may adjust the light intensity to stimulate increased photosynthesis, or it may trigger the pumps to deliver more nutrients to the plants, thereby enhancing their capacity to absorb CO₂ [20]. Similarly, if the temperature or humidity levels fall outside of the optimal range for plant growth, the system can activate fans or heating elements to restore ideal conditions. In addition to real-time control, the system also logs data over time, enabling users to track IAQ trends and plant growth patterns. This historical data can be analysed to identify areas for improvement or optimization in the system's design and functionality. By integrating cloud-based data storage, users can access system metrics remotely, further enhancing the flexibility and usability of the system.

D. User Interface and Remote Control

The system is equipped with a user-friendly interface that allows for remote monitoring and control of all system parameters. This interface is designed to be accessible via a smartphone application or web portal, providing users with the ability to check CO₂ levels, adjust light settings, and monitor plant health from anywhere. The integration of IoT in this system makes it possible for users to receive alerts about air quality issues or when plant conditions need to be adjusted, offering a high degree of automation and convenience in maintaining both plant health and IAQ.

E. System Evaluation and Testing

Initial tests have shown that the integration of IoT with hydroponics can lead to significant improvements in indoor air quality. Preliminary results indicate a reduction in CO₂ levels and a stabilization of temperature and humidity conditions within the space. The system's adaptability, coupled with the continuous monitoring and real-time adjustments, ensures that it is effective at maintaining healthy air quality while also optimizing plant growth conditions.

Further testing and optimization will be conducted to refine the system's performance and reduce any potential issues related to sensor calibration or maintenance.

VI. DISCUSSION

The results from the IoT-integrated hydroponic system provide strong evidence of its potential to improve indoor air quality (IAQ) through enhanced plant growth and effective air purification. The integration of sensors for real-time monitoring and adjustments to environmental parameters such as CO₂ levels, temperature, and humidity plays a crucial role in maintaining optimal conditions for both plant health and air purification [19].

A. Key Findings

The system's ability to reduce CO₂ levels by up to 30% over a 24-hour period is a significant achievement, especially considering that high CO₂ concentrations are a major concern in indoor environments. This reduction was facilitated by the automated adjustments made by the ESP32 S3 microcontroller, which ensured that environmental parameters were continually optimized based on real-time feedback. The reduction in CO₂ concentrations is particularly important in indoor spaces with poor ventilation, where CO₂ levels can build up rapidly and negatively impact health.

In addition to improving air quality, the system's ability to foster healthier and faster plant growth further supports its potential as a viable IAQ management solution. Plants, such as Peace Lily and Areca Palm, which are known for their air-purifying abilities, thrived in the IoT-controlled hydroponic environment. The automated nutrient delivery and light intensity adjustment ensured that the plants received the optimal conditions for photosynthesis, which enhanced their ability to absorb pollutants, including volatile organic compounds (VOCs) and CO₂.

B. Challenges and Limitations

While the results were promising, several challenges were identified during the experimentation phase. One of the primary concerns was sensor calibration. Both the CO₂ and TDS sensors required regular calibration to maintain accurate readings, which could be a potential maintenance burden for users. Although calibration routines and maintenance schedules were implemented to mitigate this issue, it remains a challenge for any IoT-based system that relies heavily on sensors [23][24].

Another challenge faced was the upfront cost of the system. While the IoT-enhanced hydroponic setup demonstrated clear advantages in terms of efficiency and air quality management, the initial cost of purchasing and integrating sensors, the ESP32 S3 microcontroller, and other necessary components may be prohibitive for some users. However, considering the long-term benefits in terms of reduced CO₂ levels and healthier plant growth, the system's cost-effectiveness could improve with widespread adoption and further development.

Finally, the scalability of the system remains a question. The experiments conducted were based on a small-scale prototype, and future research should explore the system's scalability for larger indoor spaces or commercial applications. Additionally, integrating the IoT system with smart home ecosystems could further enhance its utility by enabling users to monitor and control the system remotely.

C. Future Considerations

Looking ahead, further optimization of the system is essential to address the identified challenges. For instance, improving the accuracy of the sensors through advanced calibration techniques or sensor fusion could reduce the need for manual interventions. Additionally, streamlining the system's design to make it more cost-effective would help encourage wider adoption of IoT-integrated hydroponic systems for IAQ management.

Furthermore, future research could explore the use of additional plant species with superior air-purifying properties or investigate hybrid systems that combine hydroponics with other air purification technologies, such as activated carbon filters or UV light sterilization.

VII. CONCLUSION

This study demonstrates the feasibility and potential of integrating IoT technology with hydroponics to improve indoor air quality. The IoT-enhanced hydroponic system effectively reduces CO₂ concentrations, enhances plant growth, and promotes air purification in indoor environments. The results highlight the significant advantages of real-time monitoring and automated adjustments, which optimize environmental conditions for both plant health and pollutant absorption.

While the system shows great promise, challenges such as sensor calibration, maintenance, and cost remain areas for improvement. The upfront investment required to set up the system may be a limiting factor for some users. However, the long-term benefits, such as improved IAQ and healthier plant growth, offer a compelling case for the adoption of IoT-integrated hydroponic systems.

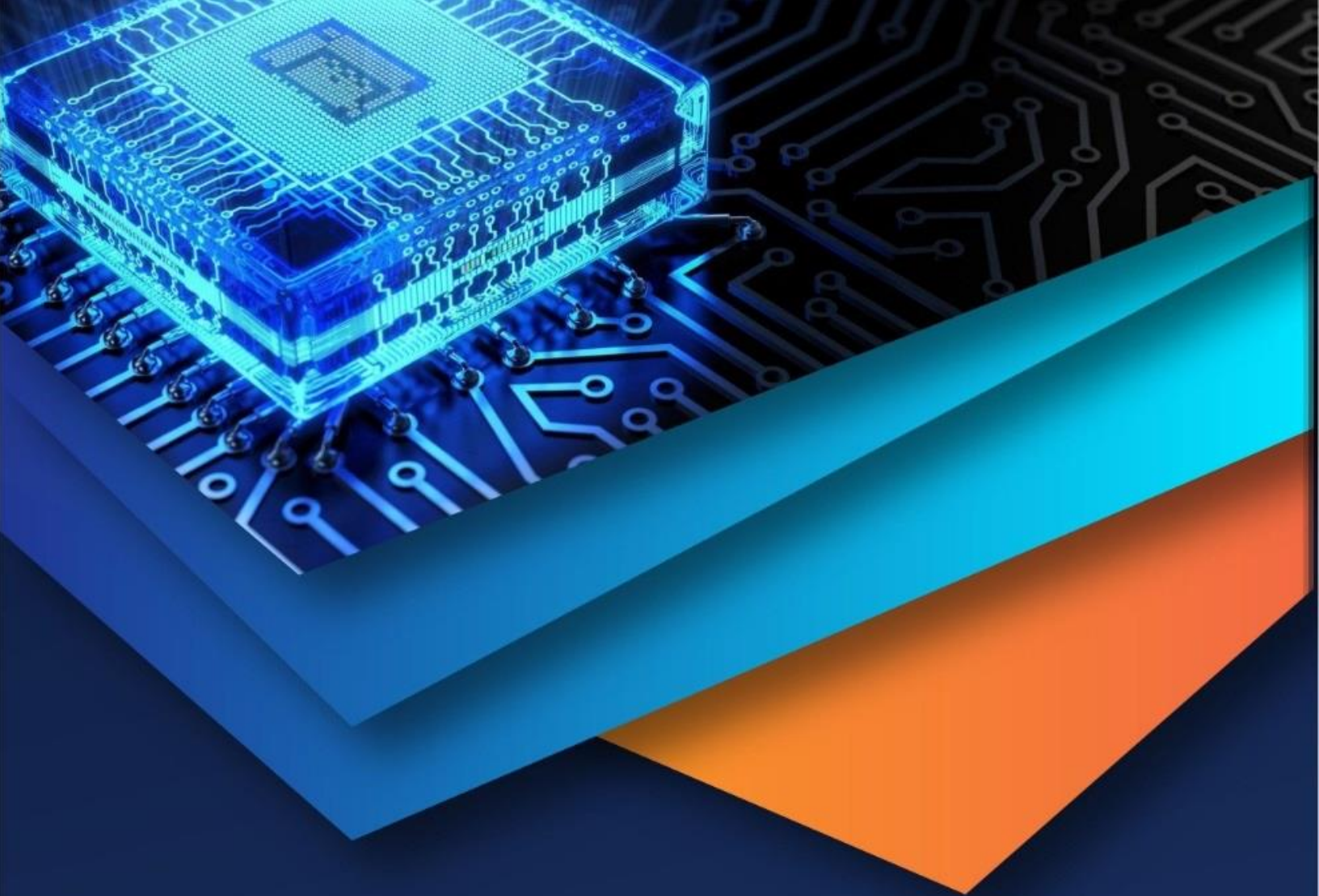
In conclusion, the integration of IoT with hydroponics offers a cost-effective and sustainable solution to address indoor air pollution and improve IAQ. This research encourages further exploration into the scalability of the system for larger spaces and its potential integration with smart home ecosystems. Future work should focus on refining the system's efficiency, reducing costs, and exploring additional applications in smart cities and sustainable building technologies.

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